

Extension of Koopmans' theorem. I. Derivation*†

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It is shown that the spin orbitals that diagonalize the generalized Hartree-Fock potentials [Int. J. Quantum Chem. Symp. 8, 501 (1974)] for a correlated wavefunction, together with the corresponding orbital energies, give a natural extension of Koopmans' theorem to correlated reference states.

I. INTRODUCTION

Koopmans' theorem¹ is widely used for approximating ionization potentials and electron affinities of atoms and molecules. Two major drawbacks to the use of this (valid) theorem are that it assumes a Hartree-Fock (HF) description (i.e., no correlation) for the parent species, and it makes no provision for relaxation of the other electrons when one is added or removed. These two deficiencies can be eliminated if one treats the correlation problem twice: once in the parent species and once in the ion. However, recent work² in our laboratory suggests that highly accurate transition energies (ionization potentials and electron affinities) may nevertheless be obtainable even if one treats the correlation problem just once (in the parent system), and then in the spirit of Koopmans' theorem treats the transition as a one-electron problem.

II. DERIVATION

The original idea behind Koopmans' theorem will be restated in a language suitable for extending the theorem so as to account for correlation and relaxation effects. There are really two problems to solve: (A) obtain a wavefunction Φ^N for the parent system, and (B) find an orbital u_r to contain an added electron if you want state r of the negative ion, or an orbital v_s to be deleted if you want state s of the positive ion:

$$\Phi_r^{N+1} = (N+1)^{-1/2} \alpha u_r(x_0) \Phi^N(x_1 \dots x_N) = \Theta_r^\dagger \Phi^N, \quad (1a)$$

$$\Phi_s^{N-1} = N^{1/2} \int v_s^*(x_0) \Phi^N(x_0 x_1 \dots x_{N-1}) dx_0 = \Theta_s^\dagger \Phi^N. \quad (1b)$$

Here α is the antisymmetrizer, $\Theta_r^\dagger = \sum_i a_i^\dagger c_{ir}^\dagger$, $\Theta_s^\dagger = \sum_j a_j \times (c_{js}^\dagger)^*$ (or $u_r = \sum_i \varphi_i c_{ir}^\dagger$, $v_s = \sum_j \varphi_j c_{js}^\dagger$), and $a_i^\dagger$ is the creation operator for the basis spinorbital φ_i .

If for problem (A) we take a normalized single determinant Φ^N , and if $w = u_r$ or v_s (an empty or occupied orbital), then the energy difference ϵ_w between the ionic and parent species, defined by

$$\epsilon_w = \frac{\langle \Theta^\dagger \Phi^N | H | \Theta^\dagger \Phi^N \rangle}{\langle \Theta^\dagger \Phi^N | \Theta^\dagger \Phi^N \rangle} - \langle \Phi^N | H | \Phi^N \rangle, \quad (2)$$

satisfies the identity $\epsilon_w = \langle w | H_0 | w \rangle$, where H_0 is the HF Hamiltonian for the reference state Φ^N and

$$H = \sum_{ij} h_{ij} a_i^\dagger a_j + \frac{1}{2} \sum_{ijkl} g_{ijkl} a_i^\dagger a_j^\dagger a_l a_k.$$

The identity $\epsilon_w = \langle w | H_0 | w \rangle$, which holds for any arbitrary orbital basis or reference state Φ^N , is sometimes (mistakenly) referred to as Koopmans' theorem. Nesbet³

has pointed out, however, that Koopmans' contribution was not this identity, which has no physical content; rather, it was the physically meaningful theorem that if Φ^N is the (variational) HF solution to problem (A), then the variational solution to problem (B) is to choose the orbitals u_r and v_s to be the HF (virtual or occupied) canonical orbitals of the parent system, for then Φ_r^{N+1} and Φ_s^{N-1} will be stable with respect to further variations in u_r or v_s .

Now ϵ_w is still an orbital energy even if correlation is included in Φ^N ; for if Φ^N is frozen and w is varied, the right side of (2) depends only on the parameters in the orbital w . Further, the form of the approximate ionic wavefunctions in (1) allows for considerable relaxation (i.e., polarization or rearrangement) effects in comparison with the HF determinant plus or minus one canonical HF orbital, if Φ^N includes correlation. Finally, ϵ_w takes a simple form if problem (A) is solved by taking Φ^N to be an eigenfunction⁴ of a model (or the exact) Hamiltonian H , for then the second term on the right of (2) can be written

$$\langle \Phi^N | H | \Phi^N \rangle = \langle \Phi^N | \Theta \Theta^\dagger H | \Phi^N \rangle / \langle \Theta^\dagger \Phi^N | \Theta^\dagger \Phi^N \rangle,$$

so that (2) becomes

$$\epsilon_w = \langle \Phi^N | \Theta [H, \Theta^\dagger] | \Phi^N \rangle / \langle \Phi^N | \Theta \Theta^\dagger | \Phi^N \rangle.$$

Using $\Theta^\dagger = \Theta_r^\dagger = \sum_j a_j^\dagger c_{jr}^\dagger$ for the case of electron capture enables us to rearrange (2) into the following matrix equation for the expansion coefficients $c_{jr}^\dagger$ of u_r :

$$\sum_{ij} c_{ir}^{\dagger*} (V_{ij}^\dagger - \epsilon_r^\dagger S_{ij}^\dagger) c_{jr}^\dagger = 0 \text{ or } \mathbf{c}^{\dagger\dagger} (\mathbf{V}^\dagger - \epsilon^\dagger \mathbf{S}^\dagger) \mathbf{c}^\dagger = 0, \quad (3)$$

where

$$V_{ij}^\dagger = \langle \Phi^N | a_i [H, a_j^\dagger] | \Phi^N \rangle \text{ and } S_{ij}^\dagger = \langle \Phi^N | a_i a_j^\dagger | \Phi^N \rangle.$$

Problem (B) is to find variationally stable solutions to (3); these are given by solving

$$\mathbf{V}^\dagger \mathbf{c}^\dagger = \mathbf{S}^\dagger \mathbf{c}^\dagger \epsilon^\dagger. \quad (4)$$

Similarly, for electron removal, we obtain $\mathbf{V}^\dagger \mathbf{c}^\dagger = \mathbf{S}^\dagger \mathbf{c}^\dagger \epsilon^\dagger$, with

$$V_{ij}^\dagger = -\langle \Phi^N | a_j^\dagger [H, a_i] | \Phi^N \rangle \text{ and } S_{ij}^\dagger = \langle \Phi^N | a_j^\dagger a_i | \Phi^N \rangle.$$

$\mathbf{V}^\dagger$ and $\mathbf{V}^\dagger$ are matrix representations of the generalized Hartree-Fock one-particle potentials^{5,6} of an eigenfunction of the Hamiltonian. Eigenfunctions to these potentials are termed natural transition orbitals. Just as the natural spin orbitals⁷ of a reference many-body state Φ^N are those orbitals whose occupation numbers $n_i = \langle \Phi^N | a_i^\dagger a_i | \Phi^N \rangle$ are variationally stable, the natural tran-

sition spin orbitals⁵ are analogously those whose orbital energies (2) are variationally stable, as seen in (4).

We therefore propose an extension of Koopmans' theorem: Given a model Hamiltonian for an N -electron system, and a correlated wavefunction Φ^N which is an eigenfunction of this model Hamiltonian, then if the orbitals u_r or v_s are natural transition orbitals (eigenvectors of the matrices $V^>$ or $V^<$ defined above), wavefunctions formed from Φ^N by adding u_r or deleting v_s have stationary energy with respect to further variations involving these orbitals.

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†While this manuscript was in its final stages of preparation, we received a letter from R. G. Parr stating that he and his colleagues had obtained a formula formally identical to ours

for obtaining ionization potentials, from a consideration of the long-range behavior of the natural spin orbitals. See M. M. Morrell, R. G. Parr, and M. Levy, J. Chem. Phys. (to be published).

¹T. Koopmans, *Physica (Utr.)* **1**, 104 (1933).

²O. W. Day, D. W. Smith, and R. C. Morrison, J. Chem. Phys. **62**, 115 (1975) (following paper).

³R. K. Nesbet, *Adv. Chem. Phys.* **9**, 321 (1965).

⁴For example, if Φ^N is approximated by a configuration-interaction wavefunction with all configurations included (a "full CI" over the orbital basis used), then the model Hamiltonian has the form shown just beneath Eq. (2), where the summations are over only the finite orbital basis being considered. Other possible choices for Φ^N would be an eigenfunction of the Hubbard Hamiltonian or the Pariser-Parr-Pople Hamiltonian, for example.

⁵O. W. Day, D. W. Smith, and C. Garrod, *Int. J. Quantum Chem. Symp.* **8**, 501 (1974).

⁶D. W. Smith and O. W. Day, *Int. J. Quantum Chem. Symp.* **8**, 511 (1974).

⁷P. O. Löwdin, *Phys. Rev.* **97**, 1474 (1955).